

# Key Challenges Faced

From design assumptions to a deployable student risk system

01

## Dataset Strategy Change

The original dual-dataset fusion idea was simplified to one consistent dataset to avoid feature mismatch and unreliable interpretation.

02

## First-Semester GPA Gap

Some first-semester students had no official GPA, so provisional academic indicators were needed without changing the model structure.

03

## Model Training Trade-off

Random Forest, XGBoost and linear models were compared because accuracy alone was insufficient; recall mattered for finding at-risk students.

04

## Email Delivery Limits

Bulk notification testing exposed SMTP rate limits and invalid test addresses, requiring controlled advisor-approved sending.

AURA

implementation focus

Final design response: reliable data, cautious prediction, interpretable output, and human control over sensitive alerts.